A Malicious and Anti-Malicious Information Propagation Dynamics Model Based on Higher-Order Diffusion Networks

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***Abstract*—As people benefit from the convenience of social networks, they are also exposed to security risks from malicious information. This article proposes a malicious and anti-malicious information propagation dynamics model based on higher-order diffusion networks to address these issues. Firstly, to tackle the problem of individuals' behavior being swayed by the emotional content of malicious information during propagation, we have established an emotional influence mechanism. At the same time, adding user preferences and external drivers proposes a way to calculate individual influence. Secondly, in response to the discrepancies in the propagation efficiency of malicious information among varying users, we establish a higher-order diffusion network and quantify these differences through the stratification of users into three distinct layers of nodes. Additionally, considering the implicit relationship among potential user groups, we establish a group interaction mechanism by mining group attributes and refine the structure of the propagation network. Lastly, given the coexistence and opposition between malicious and anti-malicious information, dynamic game theory is applied to define a state transition equation incorporating anti-malicious information propagators. Consequently, we propose the SAIR model, a novel paradigm for understanding the propagation of malicious and anti-malicious information using higher-order diffusion networks.**

***Index Terms*—Malicious information, anti-malicious information, information propagation, higher-order diffusion networks.**

I. Introduction

MALICIOUS information refers to groundless statements fabricated out of thin air and is typically spread for personal purposes through various means.With the development of society and the Internet, malicious information propagation exhibits characteristics such as faster dissemination, broader impact, greater influence, and stronger concealment compared to traditional interpersonal communication. Malicious information significantly impacts our daily lives, causing damage to personal reputations, inflicting mental distress on victims, and giving rise to issues such as a deteriorating social atmosphere, rising crime rates, and economic losses, thereby posing a substantial threat to national and public security [1], [2], [3], [4], [5], [6], [7]. Therefore, studying the trend of malicious information propagation holds great significance in guiding public opinion and managing emergencies.

Currently, research on malicious information dissemination can be broadly divided into two directions. The first direction is to establish information propagation models to analyze the trend of dissemination, such as infectious disease models, as described in literature [8], [9], [10], [11], [12], [13], [14], [15]. The second direction is to utilize deep learning algorithms to predict propagation based on certain user and network features in social media networks, as demonstrated in literature [16], [17], [18], [19].

By combining real-life experience and research, it has been found an interesting phenomenon in the spread of information on social networks. The dissemination behavior is often influenced by multiple factors, including external influences and our personal preferences. Some people tend to spread information containing positive emotions to establish their social image, while others tend to spread information containing negative emotions to vent their stress on social platforms. At the same time, it is possible that we will have different reactions when receiving information from different groups. For example, if the source of a piece of information is a field expert with numerous fans, we tend to believe it and spread it. Conversely, if a relatively unknown figure in the field suddenly posts a piece of information, it may not be very convincing [20]. In addition, based on our interests and spreading habits, it is possible that we will belong to a certain potential group, and users in the same group can influence each other's spread behavior and have tacit relationships, as shown in Fig. 1.

Based on the given situation, although research methods and technologies in the field of information dissemination are continually advancing, several challenges persist:

1) The dissemination process is influenced by malicious information that contains emotional elements. Malicious information encompasses both positive and negative emotions, affecting individual dissemination behavior. Quantifying the impact of these emotions and considering the role they play in communication is a challenge.

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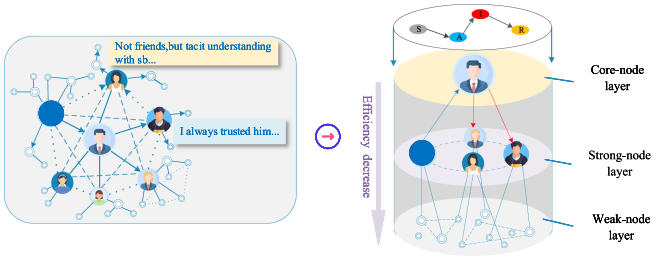


Fig. 1. The behavioral patterns of information spread and group influence in social networks.

2) Differences in dissemination efficiency exist among users. User groups within the dissemination network exhibit hierarchical patterns, and the dissemination efficiency of malicious information varies across these levels. Analyzing the influence of these differences on information dissemination is a complex issue.
3) The coexistence and antagonism between malicious and anti-malicious information pose additional challenges. Both types of information are present in the dissemination space and greatly influence user dissemination behavior. Quantifying this influence is a crucial consideration in predicting dissemination trends.

To address these challenges, This article proposes a model for the propagation dynamics of malicious and anti-malicious information based on high-order propagation networks. The model takes into account the differences in dissemination efficiency among different user groups and the game relationship between malicious and anti-malicious information. It analyzes the trend of information dissemination under a specific topic from the perspective of the emotional needs driving user dissemination behavior. The proposed model is validated using a Weibo dataset and Twitter dataset, and the results demonstrate its improved accuracy in describing the trend of malicious information dissemination on real social networks. The main contributions of this article are outlined as follows:

1) *Emotion influence mechanism:* Aiming at the problem that individual behavior is affected by the emotion contained in malicious information in the communication process, this article establishes an emotion influence mechanism. By quantifying the similarity between the emotion of malicious information and the emotion of users, it effectively elucidates the driving effect of emotions on individual propagation behavior, overcoming the limitations of traditional communication mechanisms that ignore emotional factors in propagation dynamics.
2) *High-order propagation network with latent group attributes:* In order to address the variations in dissemination efficiency among users, this article construct a higher-order diffusion network with latent group attributes and use multiple linear regression to accurately quantify the propagation efficiency differences among different user groups, breaking through the limitations of single propagation modes in existing simple models.
3) *Dynamics model for malicious and anti-malicious information propagation:* Considering the coexistence and antagonism between malicious and anti-malicious information during propagation, this article introduces evolutionary game theory to quantify the antagonism of their propagation and proposes the SAIR model to evaluate the dissemination effects of malicious and anti-malicious information, thereby providing more comprehensive strategic suggestions for decision-makers.

The remaining parts of this article are arranged as follows: Section 2 provides a review of related work; Section 3 presents the problem definition and feature extraction methods; Section 4 describes the details and formula derivation of the high-order propagation network and malicious information propagation model; Section 5 presents and analyzes the experimental simulation results; and Section 6 concludes the article.

II. Related Work

To address the challenges associated with studying malicious information propagation, researchers have developed various models that liken the propagation of malicious information to the spread of a contagious disease. Building upon epidemic models, several improved models have been proposed to analyze and understand the propagation of malicious information.

In response to the challenge of individual behavior being influenced by emotional factors during the spread of malicious information, Hatfield, Cacioppo, and Rapso [21] first proposed the theory of emotional contagion. Meel and Vishwakarma [22] explored the ecosystem of malicious information, covering aspects such as classification, spread motivation, social impact, and user perception. Liu et al. [23] analyzed various detection methods utilizing sentiment, emotion, and stance features; however, these methods are primarily applied to post-hoc detection rather than propagation dynamics modeling. Lühring et al. [24] distinguished between prior emotional states and emotional reactions to fake news, finding that fake news elicited higher levels of anger and less happiness; nonetheless, that study only explored the impact of emotions on identification capabilities and failed to establish a quantified relationship between emotional similarity and propagation behavior. Indu and Thampi [25] proposed a misinformation detection mechanism based on sentiment analysis and user behavior analysis. However, these studies did not specifically apply emotion analysis to the field of malicious information propagation, ignoring the dynamic impact of the similarity between malicious information emotions and users' own emotions on propagation probability. Building upon this, this article establishes an emotion influence mechanism to quantify the catalytic effect of emotional similarity on dissemination behavior.

There are variations in the efficiency of information propagation among different users. Barabasi [26] and Watts [27] discovered the scale-free network and small-world network, respectively, revealing two characteristics of social networks: power-law distribution and the small-world phenomenon. The structural characteristics of social networks and the interaction efficiency of complex graph data play a role in the process of information propagation [28], [29]. During information propagation, nodes with different degrees of importance will have

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varying effects on information dissemination [30]. Nodes with high degrees are more likely to promote the spread of information, while nodes with low degrees have limited influence [31]. Chen et al. [32] pointed out that nodes with higher degrees have a more significant impact on the speed of information propagation. However, most existing studies only consider the various structural characteristics of the network as influencing factors, without systematically constructing a hierarchical higher-order network with node and edge attributes that better align with real social scenarios.

To address the coexistence and antagonism of malicious information propagation, Wang et al. [33] studied the complex dynamic coupled diffusion process of positive and negative information dissemination from the perspectives of user emotions and social relationships. Zhao et al. [34] investigated the impact of official anti-malicious information on the spread of malicious information and found that the ratings and credibility of official media can significantly influence the scope of malicious information dissemination. Zhong et al. [35] proposed a rumor dissemination control and intermittent control strategy with an anti-rumor mechanism, but this study lacked quantitative modeling of the dynamic game intensity between malicious and anti-malicious information during the dissemination process. Ding et al. [36] proposed a rumor dissemination and rebuttal strategy model based on the classic SIR framework, considering the forgetting effect, but this study did not consider the social capital competition relationship between the two types of information disseminators. Although previous studies have discussed the co-dissemination of malicious and non-malicious information, they mainly focused on malicious information and did not consider the impact of the antagonistic game relationship between malicious and anti-malicious information on the dissemination trend. Therefore, this article proposes a model based on a high-order propagation network to capture the dynamics of malicious-anti-malicious information propagation.

III. Problem Definition

A. Feature Extraction and Relevant Definitions

Exploring the malicious information propagation process based on high-order propagation networks essentially involves extracting relevant features from three levels: the user itself, the user's friends, and the user's social groups, and conducting modeling and analysis. This article focuses on the propagation of malicious information and extracts relevant attributes that affect information propagation, including user attributes, the influence of same-order friends, the influence of high-order friends, and the influence of group attributes on users. Through the analysis of these factors, the driving forces behind the dynamics of information propagation can be further studied.

User Self-attributes: Whether a user participates in the spread of malicious information is closely related to the user's own attributes. This article explores key influencing factors such as emotional similarity, content fit, and individual communication enthusiasm by mining user tags and historical behaviors.

The Definition 1. Emotional similarity: Emotional similarity refers to the degree of similarity between the emotional polarity of a user's recent information dissemination and that of malicious information. The higher the similarity, the greater the likelihood of the user's dissemination. To quantify this emotional similarity, this paper constructs a set of emotion types based on an emotion dictionary, where each dimension represents a specific emotion type:

$$E = \{e_1, e_2, \ldots, e_n\} \tag{1}$$

For each historical message, an emotion vector is constructed based on the extracted sentiment words and modifiers. Each dimension of the emotion vector represents a sentiment, and the value indicates the intensity of the sentiment. The intensity of the sentiment is primarily determined by the dictionary score, frequency of occurrence, and modifiers. For a single message, its intensity I_i on the sentiment dimension e_i is calculated as follows:

$$I_i = \sum \left(Score_{\text{base}} \cdot \prod \theta_m \right), \quad m \in M \tag{2}$$

where $Score_{\text{base}}$ is the base score in the sentiment lexicon, M is the set of modifiers, and θ_m is the enhancement or attenuation coefficient associated with the modifiers.

Considering the dynamic evolution of user preferences, different weights are assigned to the vector based on temporal proximity. For the j-th piece of information published at time t_j, its weight w_j is given by the following exponential decay function:

$$\omega_j = \exp\left(-\lambda \left(t_{current} - t_j\right)\right) \tag{3}$$

where λ is the time decay constant, ensuring that information closer to the current time has a higher weight. If the integrated historical emotional vector is represented as V, and the current information emotional vector is represented as $V_{current}$, then the formula for calculating emotional similarity can be expressed as:

$$\text{Semo}\left(v_i\right) = \frac{V \cdot V_{current}}{\|V\| \cdot \|V_{current}\|} \tag{4}$$

The Definition 2. Content fit: Content fit is used to describe the matching degree between the label of malicious information and the interest label of the user. The content fitness refers to the degree of match between the user's interest tags and the tags of malicious information, indicating the likelihood of the user propagating the information. In this article, the content fitness is defined as follows:

$$Infomatch(v_i) = \frac{|Tag_{(malicious)} \cap Tag_{(user)}|}{|Tag_{(malicious)} \cup Tag_{(user)}|} \tag{5}$$

where $Tag_{(\text{malicious})}$ represents the content label of malicious information and $Tag_{(\text{user})}$ represents the hobby label of the user.

External Influence: In social networks, the probability of a user participating in information propagation is not only related to the user's own attributes, but also to the influence of their friends who participate in information propagation. Calculating the connectivity and propagation behavior similarity between a user and their friends can to some extent reflect the strength of this influence. The stronger the connectivity with friends and the

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greater the propagation similarity, the higher the probability that the user will participate in information propagation.

The Definition 3. Information popularity: Information popularity is used to describe the popularity of information. The process of information propagation goes through three stages: "rise-balance-decline". The popularity of the information will gradually decrease over time. Since the rate at which information's popularity decays is often related to its current level of attention, the higher the remaining attention, the more significant the decay process. This decay property is similar to the half-life property of physical elements. Therefore, this article introduces a half-life function to describe the decay process of the popularity of topics. The popularity of information is defined as:

$$Pop(t) = \left(\frac{1}{2}\right)^{\frac{t-t_0}{w}} \tag{6}$$

where t and t_0 represent the time of information propagation and the time of information appearance, respectively, and w represents the regularization factor or information propagation period.

The Definition 4. User Connectivity: User connectivity is used to describe the degree of closeness between a user and their friends in a social network. The more common friends they have, the more similar their information preferences are, and the stronger their user connectivity is. This makes it more likely for users to propagate malicious information. The formula for calculating user connectivity is defined as follows:

$$co(v_i) = simtag(v_i, v_j) + simfriends(v_i, v_j) \tag{7}$$

where $simtag(v_i, v_j)$ represents the similarity of tags between two users, and $simfriends(v_i, v_j)$ represents the number of mutual friends.

The Definition 5. Friend propagation power: Friend propagation power is defined as the degree to which the information propagated by a user's friends is viewed, commented on, and shared by others. A higher value indicates that the user's friends have a stronger driving force and the user is more likely to be influenced and engage in propagation. Based on the user's historical data, the formula is defined as follows:

$$Fridrive(v_i) = \frac{1}{N}\sum_{j=0}^{j=N}(Ns[c(v_j)] + Ns[r(v_j)] + \mu * Ns[rd(v_j)]) \tag{8}$$

where $Ns[c(v_j)]$, $Ns[r(v_j)]$, $Ns[rd(v_j)]$ represent the average number of comments, retweets, and reads per message of a friend, respectively, while μ is the weakening factor.

B. Problem Formalization

The purpose of this article is to investigate the propagation patterns of malicious and anti-malicious information in high-order propagation networks with group attributes, based on user behavior data $A = \{(a, v_i, t)\}$ on social platforms. Firstly, the core nodes, strong nodes, and weak nodes in the three-layer network and user groups are identified. Then, by mining the key features of the information itself, user friends, and potential user groups, the influence is calculated as the basis for the probability of state transition of user v_i, and the user's state at time t is predicted as $D(v_i, t) = S, A, I, R$, as well as the overall trend of topic propagation.

Problem Input:

1) The user relationship network $G = (V, E)$ in the process of malicious information propagation, where users are viewed as nodes, and following relationships between users are viewed as directed edges, forming a directed graph G.
2) The historical behavior data of users is denoted as $A = \{(a, v_i, t)\}$, which is the set of users' spreading and approval behaviors for different topics.
3) Malicious information text content $C = \{(id, text)\}$, which is the malicious information ID corresponds to the specific text content. assemble

Problem Output: Based on the above input, the output is as follows:

1) Driving force of information dissemination $Drif_1(v_i)$: This article extracts relevant factors at the user and friend level, combines them with group attributes, and uses multiple linear regression to construct the propagation influence of different information types. It also uses dynamic game theory to quantify the driving force of information propagation, denoted as $Drif_1(v_i)$.
2) Data on the spread of malicious information $S(t), I(t), A(t), R(t)$: This article introduces the SAIR model to construct the anti-malicious information state, and combines the user's information propagation driving force $Drif_1(v_i)$ with the model to provide theoretical basis for the infection rate and removal rate in the model. Then, using the binomial distribution, the infection rates $\delta_1(t)$, $\delta_2(t)$ of malicious and anti-malicious information are calculated, as well as the corresponding removal rates $1 - \delta_1(t), 1 - \delta_2(t)$. Finally, based on mean field theory, a differential dynamic model is constructed to calculate the user sets $S(t), I(t), A(t), R(t)$ at different time periods and simulate the information propagation trend.

IV. Model

A. Model Framework

To solve the above problem, the model framework constructed is shown in Fig. 2. First, by extracting the content attributes of malicious information, the historical behavior attributes of users, and the structural attributes of social networks, a high-order propagation network is constructed to quantify the multidimensional influence of malicious and anti-malicious information. Second, using multidimensional influence as the core input of evolutionary game theory, the competitive landscape of malicious and anti-malicious information at the user's psychological level is quantified, thereby generating the propagation driving force of each type of information. Then, the propagation driving force is mapped to state transition probability to determine the possibility that a user will be transformed into an "infected state" or an "anti-infected state" after being exposed to information at

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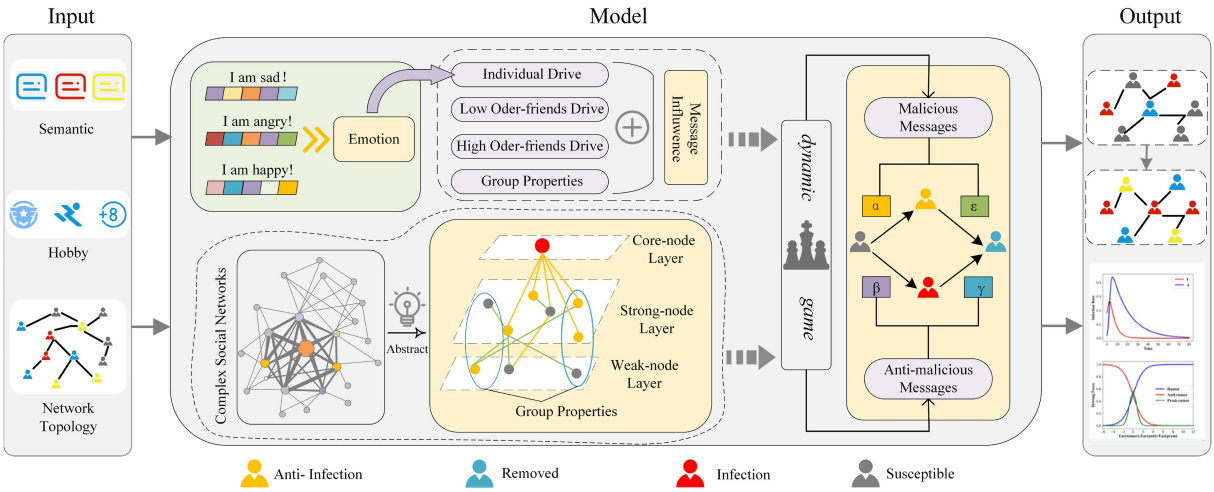


Fig. 2. Model framework.

a specific time. Finally, using the SAIR differential equation that includes the anti-infected state, the dynamic transition probability is integrated into the overall evolutionary process, ultimately presenting a complete propagation landscape of malicious and anti-malicious information.

B. Model Refinement

Construction of High - level Communication Network Based on Group Attribute: Malicious information spreads through user friend relationships in social networks, but the propagation efficiency of different relationships between users varies. To quantify this difference, this article constructs a high-order propagation network from low order to high order. Meanwhile, the indicators for predicting user propagation behavior not only include user's own propagation behavior and user friend's propagation behavior, but also define potential group attributes from the individual level to the group level. Firstly, based on the Pagerank algorithm, the core nodes with key propagation roles are extracted, and the connection relationship between core nodes and ordinary nodes is examined to divide the network into three layers: core-node strong-node weak-node. Secondly, using propagation behavior similarity as the input of KNN algorithm, users without direct contacts are clustered into different groups, and the preferences and activity levels of groups are extracted to examine their influence on propagation behavior.

To divide users into groups, we first calculate the propagation behavior similarity $Prb(v_i, v_j)$, defined as the number of times users v_i and v_j perform the same propagation behavior for a certain piece of information. Even without direct connections, if the propagation behavior similarity is high enough, users are considered to have potential group relationships. When studying the same amount of information, the more times users propagate the same information, the higher their intimacy and the greater their promotion effect on propagation. The formula for propagation behavior similarity is defined as follows:

$$Prb(v_i, v_j) = |(a, v_{ij}) \in (D_{v_i} \bigcap D_{v_j})| \tag{9}$$

where D_{v_i} represents the historical behavior dataset of user, represents the historical behavior dataset of user v_j, and (a, v_{ij}) denotes the same reaction made by users v_i and v_j to the same information.

Group activity in group attributes refers to the historical information propagation of the group to which the user belongs. The more frequent the group spreads information, the more active the user's belonging group is. Users belonging to groups with higher group activity are more likely to participate in information propagation. The definition of group activity in this article is as follows:

$$Gact(g_i) = \sigma \cdot Nbs[origmes(v_i)] + \gamma \cdot Nbs[partmes(v_i)] \tag{10}$$

In which, $Gact(g_i)$ represents the group g_i activity level, indicating the amount of original and propagated information created and disseminated by group in a period of time prior to the occurrence of malicious information. $Nbs[origmes(v_i)]$ represents the total amount of original information created by group during this period, while $Nbs[partmes(v_i)]$ represents the total amount of propagated information disseminated by group g_i. σ and γ are the attenuation factors.

Group preference ($Gpre(g_i)$) is used to describe the degree of similarity between a group's propagation preference and the malicious information labels. In this study, we collected and organized the information propagated by group one month prior to the outbreak of malicious information. We used the Textrank algorithm to extract keywords and then normalized the calculations using the Jaccard coefficient. A higher Jaccard coefficient indicates a greater degree of alignment between the group's propagation preferences and the current malicious information, and users in the group are more likely to engage in propagation behavior. The group preference is defined as follows:

$$Gpre(g_i) = \left| \frac{Gtag(g_i) \bigcap Tag(malicious)}{Gtag(g_i) \bigcup Tag(malicious)} \right| \tag{11}$$

where $Gtag(g_i)$ represents the propagation keywords of the group and represents the malicious information keywords mentioned earlier.

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Malicious - anti-malicious Information Transmission Driving Force Analysis: In social networks, the successful propagation of a piece of information is influenced by multiple factors and dimensions. At the individual level, the impact of information dissemination is influenced by the user's own attributes, such as emotional similarity, individual initiative, and content alignment. At the group level, the impact is influenced by group attributes, such as group similarity and group activity. At a lower dimension, the propagation behavior of friends can enhance the impact of information, while at a higher dimension, information propagated by higher-order nodes is more easily accepted and has higher propagation efficiency. These factors are relatively independent at the cognitive level, and their effects can be linearly additive. In this study, user's own attributes, the influence of lower-order nodes, the influence of higher-order nodes, and group attributes correspond to different psychological mechanisms, and their roles in communication decisions are relatively independent. Furthermore, linear models have significant advantages in statistical testing capability and computational efficiency. Therefore, we adopt a multiple linear regression framework to define multidimensional information influence:

$$\begin{aligned} Multeff(v_i) = {} & \gamma_0 + \gamma_1 * Useratt(v_i) \\ & + \gamma_2 * lnodeinf(v_i, n_l) \\ & + \gamma_3 * hnodeinf(v_i, n_h) \\ & + \gamma_4 * Groupatt(v_i) \end{aligned} \tag{12}$$

where $Useratt(v_i)$ represents the user's own attributes, $lnodeinf(v_i, n_l)$ represents the influence of low-order nodes, $hnodeinf(v_i, n_h)$ represents the influence of high-order nodes, and $Groupatt(v_i)$ represents group attributes. $Useratt(v_i)$ is defined as follows:

$$Useratt(v_i) = Semo(v_i) * Infomatch(v_i) * Pop(t) \tag{13}$$

$hnodeinf(v_i, n_h)$ is defined as follows:

$$hnodeinf(v_i, n_h) = co(v_i, v_j) * Fridrive(v_i) * (1 - r) \tag{14}$$

Similarly, low-order node influence represents the influence of information from weak nodes to strong nodes or from strong nodes to core nodes, and has a similar influence calculation method as high-order node influence. The difference is that the sources of information propagation for the two are different, thus reducing the efficiency of propagation.

$$lnodeinf(v_i, n_l) = c_o(v_i, v_j) * Fridrive(v_i) * (1 - r) \tag{15}$$

$Groupatt(v_i)$ is defined as follows:

$$Groupatt(v_i) = Gact(g_i) * Gpre(g_i) \tag{16}$$

where $\gamma_1, \gamma_2, \gamma_3$ and γ_4 represent the partial regression coefficients obtained by fitting a multiple linear regression model, reflecting the weights of self-attributes, low-order node influence, high-order node influence, and group attributes in information influence.

In high-order propagation networks, there are significant differences in information propagation efficiency between nodes at different levels. To characterize this propagation efficiency attenuation introduced by differences in network hierarchy, this paper introduces an efficiency factor r to quantify the differences in propagation efficiency. In real-world social networks, nodes with higher global influence tend to gain more trust, and the information they publish or forward experiences less attenuation during propagation. Based on this phenomenon, this paper uses Pagerank score as an approximate measure of a node's global structural influence and combines it with the degree of information content matching to construct a comprehensive propagation efficiency score for nodes:

$$mpr(v_i) = \tau * [k_1 * pr(v_i) + k_2 * Infomatch(v_i)] \tag{17}$$

Social networks are complex network environments where malicious and anti-malicious information often coexist. When users face these two types of information, their self-interest and conformity psychology drive people to choose information that spreads more widely and brings higher benefits. Therefore, evolutionary game theory is introduced to obtain the malicious information propagation driving force of user v_i, denoted as $Drif_1(v_i)$:

$$Drif_1(v_i) = \frac{e^{earn_1 - earn_2}}{1 + e^{earn_1 - earn_2}} \tag{18}$$

where $earn_1$ and $earn_2$ represent the social capital gain functions for spreading malicious and anti-malicious information, respectively.

Assuming that the proportion of users in v_i's friend network who spread malicious information is P_1, the proportion of friends who spread anti-malicious information is $1 - P_1$. Although there may exist friends who do not forward any information, their forwarding strategy does not affect the user's strategy. This article believes that conformist behavior can gain some social capital in the short term, and the level of information influence also affects the user's judgment of social capital gain. Therefore, the definitions of $earn_1$ and $earn_2$ are as follows:

$$earn_1(v_i) = P_1 * Multeff(v_i) \tag{19}$$

$$earn_2(v_i) = (1 - P_1) * Multeff(v_i) \tag{20}$$

Similarly, it is possible to obtain the drive force of user for spreading anti-malicious information, denoted as $Drif_2(v_i)$:

$$Drif_2(v_i) = \frac{e^{earn_2 - earn_1}}{1 + e^{earn_2 - earn_1}} \tag{21}$$

Malicious-anti-malicious Information Propagation Model: In real-world social media public opinion, anti-malicious information often appears simultaneously with malicious information and creates a confrontation. Traditional SIR models only simulate the spread of a single message, neglecting this two-way confrontation. Considering the inhibitory effect of anti-malicious information on the spread of malicious information in real social networks, this paper introduces an anti-malicious information state (Anti-infection: referring to the user group spreading anti-malicious information) into the traditional SIR model to capture the dynamic game process between malicious

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and anti-malicious information. The model includes four states: State S (Susceptible): users who have not yet been exposed to malicious or anti-malicious information; State I (Infection): users who choose to spread malicious information; State A (Anti-infection): users who choose to spread anti-malicious information; and State R (Removed): users who have been exposed to both types of messages and have chosen to ignore them. Compared to the traditional SIR model, the introduction of the anti-malicious information state can quantify users' subjective decision-making process based on social capital gains, simulate the positive inhibitory effect of anti-malicious information on malicious information, and more accurately predict infection peaks and the overall lifecycle of events.

The propagation model defined in this article is based on the following three assumptions:

1) Due to the explosive nature and short duration of malicious information propagation, this article assumes that the total number of people participating in information propagation remains basically unchanged during the propagation period, i.e., $S + A + I + R = 1$.
2) Meanwhile, malicious information is transmitted through contact, and as long as a new user comes into contact with malicious information, they will have a certain infection rate.
3) The malicious information does not exist permanently and its lifecycle is set to 12 hours from the moment a user first comes into contact with it.

Based on the assumptions mentioned above, the rules for malicious information propagation in social networks are defined as follows:

Susceptible nodes, malicious information nodes, and anti-malicious information nodes interact with each other with probabilities α and β, respectively, and are transformed into malicious or anti-malicious information nodes. Since some nodes do not participate in information dissemination from start to finish, $\alpha + \beta < 1$.

Furthermore, since the information propagation has a certain lifespan, malicious and anti-malicious information nodes will transition to the removed state after a period of time. This article assumes that the transition probabilities are γ and ε, respectively, and that $\alpha, \beta, \gamma, \varepsilon \in (0, 1)$.

The following dynamic equations can be obtained based on the above rules:

$$\begin{cases} \frac{dS}{dt} = -\alpha A(t)S(t) - \beta I(t)S(t) \\ \frac{dA}{dt} = \alpha A(t)S(t) - \varepsilon A(t) \\ \frac{dI}{dt} = \beta I(t)S(t) - \gamma I(t) \\ \frac{dR}{dt} = \varepsilon A(t) + \gamma I(t) \\ S + A + I + R = 1 \end{cases} \tag{22}$$

Information dissemination is unidirectional, and the state transition of users is also unidirectional. Users change from susceptible state to malicious information dissemination state or anti-malicious information dissemination state, and finally become removed state as time goes by. Assuming that a user has m neighbors, the probability that k of these neighbors participate in malicious information dissemination follows a binomial distribution:

$$P(X = k) = \binom{m}{k} Drif_1(v_i)^k (1 - Drif_1(v_i))^{m-k} \quad (k = 0, 1, 2, \ldots, m) \tag{23}$$

The probability of any user participating in the spread of malicious information at the moment is:

$$\delta_1(t) = \sum_{k=0}^{m} \frac{k}{m} \binom{m}{k} Drif_1(v_i)^k (1 - Drif_1(v_i))^{m-k} \quad (k = 0, 1, 2, \ldots, m) \tag{24}$$

Similarly, the probability that any user can participate in the dissemination of anti-malicious information at the moment t is:

$$\delta_2(t) = \sum_{k=0}^{m} \frac{k}{m} \binom{m}{k} Drif_2(v_i)^k (1 - Drif_2(v_i))^{m-k} \quad (k = 0, 1, 2, \ldots, m) \tag{25}$$

Combine the mean field equations to obtain the kinetic equations:

$$\begin{cases} \frac{dS}{dt} = -\overline{\delta_1(t)}A(t)S(t) - \overline{\delta_2(t)}I(t)S(t) \\ \frac{dA}{dt} = \overline{\delta_1(t)}A(t)S(t) - (1 - \overline{\delta_1(t)})A(t) \\ \frac{dI}{dt} = \overline{\delta_2(t)}I(t)S(t) - (1 - \overline{\delta_2(t)})I(t) \\ \frac{dR}{dt} = (1 - \overline{\delta_1(t)})A(t) + (1 - \overline{\delta_2(t)})I(t) \\ S + A + I + R = 1 \end{cases} \tag{26}$$

C. Model Algorithm

This article abstracts a high-order propagation network with group attributes based on a real social network. Secondly, it calculates the key propagation parameters and quantifies user propagation behavior based on the driving force of information propagation. Finally, a propagation dynamics model is constructed by adding the anti-malicious information state to the traditional SIR model. The model algorithm is as follows:

The proposed model algorithm first establishes a high-order propagation network with group attributes using the KNN and Pagerank algorithms. Then, different user groups $S(0)$, $A(0)$, $I(0)$ and $R(0)$ are initialized with various states, and multiple linear regressions and evolutionary games are performed to calculate the propagation driving force of user information, which is used as the basis for user transformation. The key parameters of the model are obtained by fitting real data. Moreover, the time complexity of the KNN algorithm is $T_{KNN} = O(N)$, the time complexity of measuring information influence using multiple linear regression is $T_{inf} = O(N)$, the time complexity of calculating user driving force in evolutionary games is $T_{dir}(N)$, and the time complexity of executing the information propagation model is 1. Therefore, the time complexity of the algorithm is $T = T_{KNN} + T_{inf} + T_{dir} \sim O(N)$.

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Algorithm 1: Information Propagation Model Algorithm.

INPUT
- Propagation network:$G = \{V, E\}$
- User behavior set:$A = \{(a, v_i, t)\}$
- Topic context:$C = \{(id, text)\}$

TRAIN
- Transfer KNN algorithm to classified groups
- Transfer Pagerank algorithm to classified node layers
- Initial S, A, I, R
- for each user v_i do
  - Calculate $Semo(v_i), Infomatch(vi), Initiative(v_i)$
  - Calculate $Co(v_i, v_j), Fridrive(v_i)$
  - Calculate $Prb(v_i, v_j), Gact(g_i), Gpre(g_i)$
  - Perform multiple linear regression to calculate information influence $Multeff(v_i)$
  - Obtain parameter $\gamma_0, \gamma_1, \gamma_2, \gamma_3, \gamma_4$
  - Perform dynamic game to calculate driving force $Drif_1(v_i), Drif_2(v_i)$
- end for
- execute mean field theory from (23)

OUTPUT
- Infection rate:$\delta_1(t), \delta_2(t)$
- Immunization rate:$1 - \delta_1(t), 1 - \delta_2(t)$
- User sets:$\{S(t), A(t), I(t), R(t)\}$

V. Experiment and Discuss

A. Data Description

This article investigates the propagation of malicious information using two publicly available datasets: the Sina Weibo dataset [37] and the Twitter dataset [38], [39]. Sina Weibo is the largest social media platform in China, while Twitter is the most widely used social media platform globally. Furthermore, different types of malicious information exhibit significant differences in terms of audience, impact, and eradication time. Therefore, this article selects four malicious propagation processes (referred to as malicious propagation processes for convenience in the following text) for exploration. The article conducts research on two malicious information propagation processes selected from Sina Weibo and Twitter, respectively. The processes are A: "2012 end of the world", B: "Zhou Kehua was not killed", C: "Germanwings plane crash", and D: "Sydney Siege incident".The experimental data are shown in Table I.

B. Experimental Results and Discussion

The experimental simulation in this article is divided into three parts. The first part is to conduct experimental simulation on the proposed model and compare it with the real data propagation trend to test the effectiveness of the model. The SAIR model is compared with the SIRH model to verify the effectiveness of the proposed anti-malicious information state. The second part analyzes the influence of user self-attributes and external factors on user propagation behavior, and further analyzes the necessity of various extracted features in characterizing information influence. The third part explores the influence mechanism of the high-order propagation network with group attributes on user propagation behavior and analyzes the rationality of the simulated network structure. The fourth part analyzes the relationship and development trend between malicious information and anti-malicious information from the perspective of the number of information propagations per unit time.

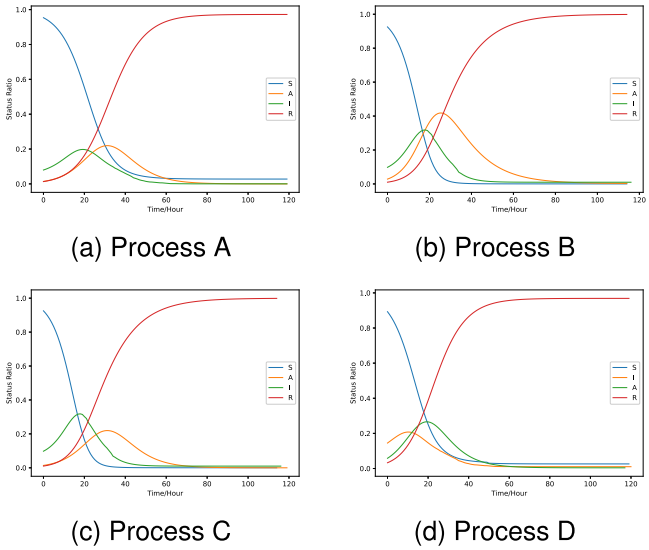


Fig. 3. Model renderings of different process.

Model Validation: This article selects four active malicious dissemination processes and models their propagation patterns during their peak periods. Each hour's propagation volume is used to segment the time periods, without considering the case of the same user participating in the dissemination of multiple malicious information topics. Additionally, it is assumed that if a participant in the malicious information topic ceases dissemination for more than 24 hours, the user is considered to be in a removed state. Fig. 2 illustrates the simulated results of the four processes' propagation trends using the model proposed in this article, along with a comparison between the model's trend and the real data.

Fig. 3(a), (b), (c) and (d) show the proportion of users in the susceptible (S), anti-malicious-information propagation (A), malicious-information propagation (I), and removed (R) states, respectively, in each time interval. It is evident from the figure that the number of users in the susceptible state continuously decreases and eventually stabilizes. Users in the A and I states exhibit similar trends of first increasing and then decreasing in the same time period. Meanwhile, the I state always reaches its peak faster and begins to decline after the proportion of users in the A state increases, which is consistent with the role of anti-malicious-information propagation in curbing malicious information in reality. The proportion of users in the removed state continuously increases and eventually stabilizes, indicating the end of the process lifecycle. Fig. 4(a), (b), (c) and (d) show the simulation trends of anti-malicious-information propagators and malicious propagators and the comparison with the actual

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TABLE I
DATA DESCRIPTION

Item	Process A		Process B		Process C		Process D	
	Malicious	Anti-malicious	Malicious	Anti-malicious	Malicious	Anti-malicious	Malicious	Anti-malicious
Number of users	821	813	1633	2258	1090	1049	3109	2679
Number of fans	200249	22443	739070	1026011	305729	345060	746672	699219
Number of edges	251549	275068	1009541	1391126	380842	464092	982151	914632
Number of behaviors	2208345	3010576	903234	3802334	30902345	10504218	9138724	6013283

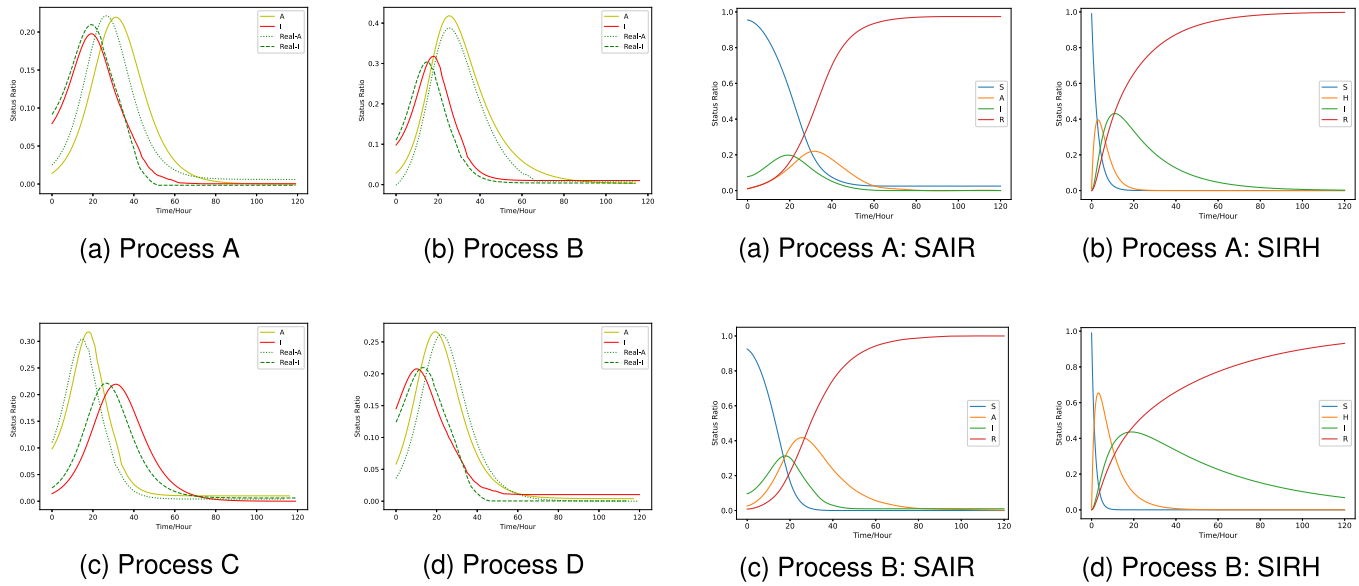


Fig. 4. Comparison of A, I states and true values in different process.

Fig. 5. Comparison of the SAIR model with the SIRH model.

trends. It can be observed that the predicted trends of the model are generally consistent with the actual trends.

To improve the sufficiency of model validation, this study introduced the classic improved information dissemination model SIRH for comparative analysis [40]. This model is an information dissemination model based on rumors, counter-rumors, stimulating rumors, and a three-way cognitive game, introducing "hyperstates" and "stimulating rumors" to observe dissemination trends by simulating the competition and cooperation among multiple parties. The comparative experiment selected two representative malicious information dissemination processes: Process A ("2012 end of the world") and Process B ("Zhou Kehua was not killed").

Fig. 5(a), (b), (c) and (d) show the experimental results of the simulated propagation curves of the SAIR and SIRH models in Process A and Process B, and is used to illustrate the suppression effect of introducing anti-malicious information state on the intensity and peak of malicious propagation. In the SIRH model, state H rapidly peaks in the early stages of propagation for both topics (3-10 hours) and then decays quickly (0.65 for topic B, 0.4 for topic A). This brief burst of activity drives state I to a relatively high peak (0.44 for both) in the subsequent period (20 hours). The SAIR model, however, exhibits a completely different propagation pattern: state A peaks around 30 hours (0.22 for topic A, 0.42 for topic B), with a significantly longer duration than state H in the SIRH model. The peak value of state I in the SAIR model is significantly lower than in the SIRH model (0.2 for topic A, 0.32 for topic B), clearly reflecting the inhibitory effect of anti-malicious information on the propagation of malicious information.

These results clearly validate the adversarial mechanism of the malicious-anti-malicious information game in the SAIR model. Compared to the stimulus-rumor unidirectional catalytic mechanism of the SIRH model, it can more accurately control the intensity of malicious information propagation. Furthermore, the introduction of higher-order propagation networks and emotional influence mechanisms further validates the rationality and effectiveness of the propagation driving mechanism and dynamic game theory proposed in this paper.

Analysis of the Mechanism of Propagation Drivers: 1) The Influence of Self and External Factors on Information Diffusion.

The purpose of this section is to study the impact weights of self-factors and external factors on user dissemination behavior. The active period of malicious information dissemination is sliced into time intervals of ten hours, and the weights of internal and external factors in the driving force are calculated and plotted in Fig. 6.

As shown in the figure, the influence of internal and external factors on the driving force of propagation varies at different time periods during malicious information dissemination. In the early stages of propagation, the weight of self-influence factors is slightly higher, but as the propagation progresses, the

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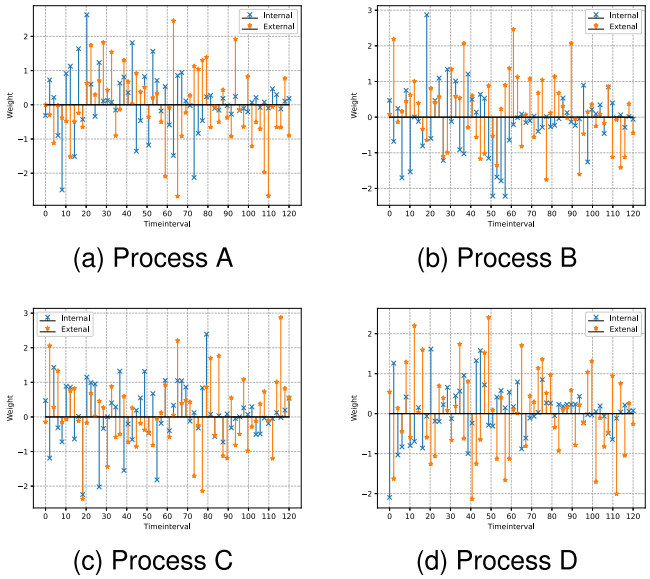


Fig. 6. Weights of self-factors versus external factors.

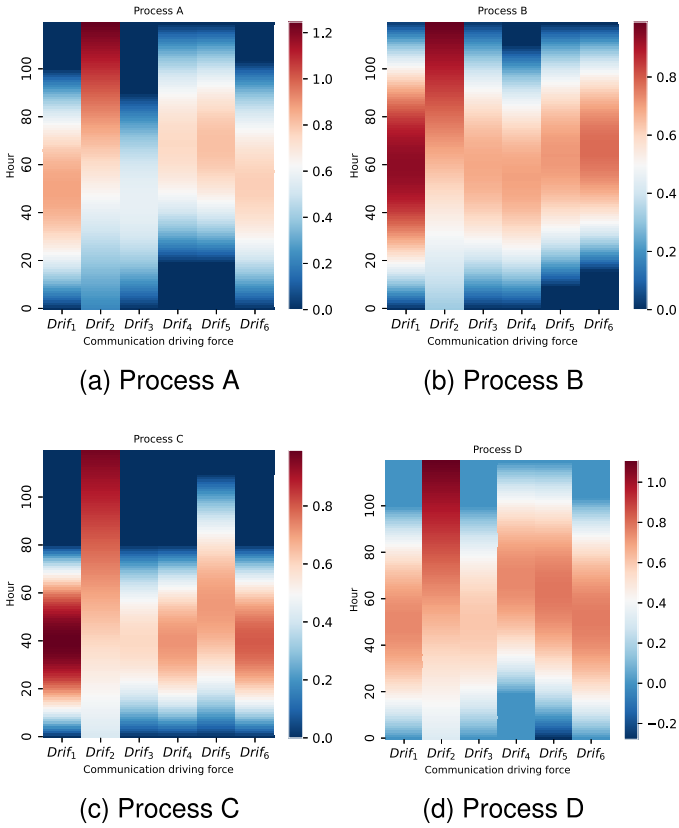


Fig. 7. Correlation of different characteristics with propagation drivers.

proportion of self-factors decreases and external factors take the dominant position. This pattern is consistent with the rules of social network propagation in reality. In the early stages of malicious information propagation, users' propagation behavior mainly depends on their own preferences. However, in the later stages, due to the strengthening of external driving factors and users' conformity psychology, propagation behavior relies more on user interaction to drive.

2) Refine the influence of characteristic factors on the driving force of propagation.

This section aims to investigate the relationship between feature factors and propagation driving forces from two perspectives: user intrinsic properties and external influence. By controlling variables, the correlation between different features and propagation driving forces was analyzed, and the results are shown in Fig. 7.

In the figure above, each column represents the trend of the malicious information propagation driving force over time with one driving factor removed while keeping all other feature factors constant. $Drif_1$ represents the trend of the propagation driving force under complete driving factors, showing a trend of first increasing and then decreasing over time, which conforms to the objective law of malicious information propagation. The dark area in process A between 40 and 60 hours represents the peak of the driving force of malicious information propagation. At this point, without considering the counteraction of anti-malicious information, the rate of propagation reaches its peak, and the proportion of the status of malicious information propagators also reaches its maximum value.

$Drif_2$ represents the trend of the propagation driving force after removing the influence of information popularity. From the experimental results, it can be observed that the starting point of the driving force is not zero, and it has been steadily increasing even after 100 hours without reaching the peak. Therefore, it can be concluded that once the constraint of information popularity is lost, the driving force will not reach its peak and will keep increasing, which does not conform to the objective stages of the "rise-balance-decline" process of information propagation. $Drif_3$ shows the trend of the driving force after removing the influence of emotional similarity. It can be clearly seen that the peak value of the driving force is much lower than the normal level. Therefore, the proposed mechanism of emotional influence in this study is of great significance and is a key indicator for simulating the process of propagation.

$Drif_4$, $Drif_5$ and $Drif_6$ represent the changes in the driving force of malicious information propagation after removing the factors of content fit, connectivity, and friend influence, respectively. As shown in the figure, the peak value of the driving force decreases and the time of its occurrence is delayed, but the overall trend of change is consistent with the true trend. Analysis shows that since all three driving features are derived from user historical data, they are not easily changed and mostly affect the strength of propagation, but will not change the overall trend of propagation.

Based on the analysis of the above experimental results, we can evaluate the hierarchical contribution of the core subsystems to the model's accuracy. First, the experimental results show that the removal of emotional similarity leads to a significant reduction in the peak value of the model's driving force, proving that the emotional influence mechanism plays a core role in simulating the intensity of malicious information outbreaks. Second, although the removal of connectivity and friend propagation power does not change the basic evolutionary trajectory,

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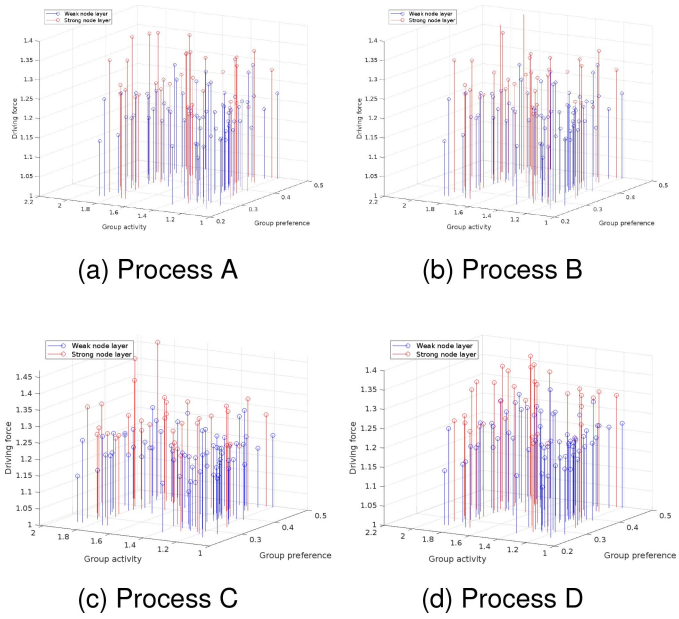


Fig. 8. Illustration of strong and weak node layers' propagation forces.

the resulting peak bias and time delay indicate that higher-order propagation networks and group attributes significantly improve the model's accuracy in quantitatively predicting the scale of propagation. The evolutionary game mechanism, as the logical hub, realizes the dynamic coupling between various feature factors, enabling the model to accurately depict the long-term competitive situation between malicious and anti-malicious information, effectively ensuring the model's robustness in complex social environments.

The Impact Mechanism of High-order Network and Group Attributes on Propagation Driving Force: The focus of this section is to examine the difference in propagation efficiency between the strong and weak node layers, as well as the mechanism by which group attributes affect propagation behavior. We selected time nodes from each active period of the propagation process, and captured snapshots of the malicious information propagation state. The real-time data of the malicious information propagation driving force, potential group activity, and potential group preference for each node were plotted in a three-dimensional coordinate system. At the same time, the data of the strong node layer was plotted in red, and the data of the weak node layer was plotted in blue. The experimental results are shown in Fig. 8.

As expected, the driving force of the strong node layer is generally higher than that of the weak node layer, and the propagation driving force also tends to increase with the growth of group attributes. Due to the existence of the efficiency factor, in the interval where the group preference and group activity are similar, the propagation driving force of the strong node layer is generally higher than that of the weak node layer. This phenomenon is related to the fact that in real social networks, users lack the ability to distinguish malicious information posted by experts or influencers they follow. In order to pursue clicks and traffic, some influencers may spread malicious information with strong emotional color, and package them as "real and

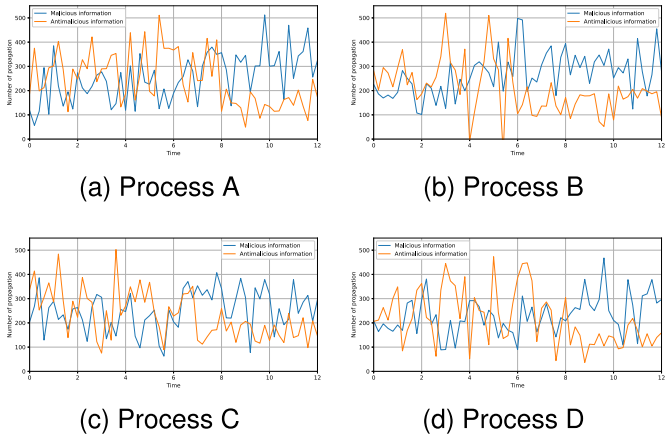


Fig. 9. Illustration of information propagation frequency over time segments.

trustworthy", and users who follow them are easily guided into becoming part of the spread of malicious information.

In the above figure, the propagation drive is positively correlated with the x and y values, and reaches its peak when the group activity and preference values reach their maximum values. This indicates that in the process of malicious information propagation, user behavior is not only directly influenced by friend relationships but is also closely related to potential group relationships. When users belong to highly active groups, i.e., users are highly active in spreading information, it can also motivate them to engage in spreading behavior. If the spreading preference of the user's group coincides with the content of malicious information, the likelihood of user spreading may also increase significantly.

In summary, this article establishes a high-order propagation network propagation model with group attributes, which is more realistic compared to traditional network models.

The Exploration of The Relationship Between Malicious and Anti-malicious Information: In this section, the propagation behavior of each process is divided into time segments, and the number of malicious and anti-malicious transmissions is plotted on the coordinate axis for each time period, in order to explore the relationship between malicious and anti-malicious information propagation. The results are shown in Fig. 9.

As shown in the above process, the high-frequency dissemination of malicious information is accompanied by an increase in the spread of anti-malicious information, and vice versa. The trend of malicious information decreases over time, ultimately dropping to a very low level. On the other hand, the spread of anti-malicious information shows an upward trend over time, ultimately surpassing malicious information in overall spread and maintaining a longer dissemination time. Through analysis combined with real social networks, we can learn that this phenomenon is closely related to the game between malicious and anti-malicious information. In the early stages of information dissemination, these two types of information have a certain cooperative relationship, and the popularity of one type of information can also drive the dissemination of opposing information. This is because after initially receiving information from a certain party, individuals have a deeper impression

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of the relevant information, but still lack the ability to identify the real information, which leads to blind communication behavior.

In addition, malicious information has a greater advantage in spreading in the early stages, which verifies the social phenomenon of "good news does not go out, but bad news spreads fast". Malicious information often has the characteristic of exaggerated language and strong emotional expression. Choosing to spread malicious information may allow users to gain short-term reputation and attention. However, as the influence of anti-malicious information expands, the malicious information will eventually be forgotten and disappear, while the anti-malicious information can continue to influence users and maintain a considerable period of time.

In summary, in the long run, the game of information dissemination process suggests that choosing to spread anti-malicious information is the optimal solution for users. This not only helps them win personal social capital but also contributes to maintaining social stability.

VI. Conclusion

In the study of social networks, we have observed a hierarchical phenomenon in user propagation efficiency, where potential groups exert a profound influence on user propagation behavior. Building upon this observation, this article proposes an emotion influence mechanism, constructs a high-order propagation network with group attributes, refines the propagation structure, and ultimately establishes a dynamic model for the propagation of malicious and anti-malicious information. The experiments demonstrate that the model results align more closely with the actual social network propagation process, effectively revealing the patterns of malicious information dissemination.

Nevertheless, this paper still has some limitations due to constraints in data and computing resources. First, the model primarily focuses on textual semantic features, while malicious information in reality is increasingly multimodal; future work will consider introducing multimodal learning to handle text, image, and video data. Second, the current analysis is based on relatively static network snapshots; to improve this, temporal graph neural networks could be introduced to simulate the dynamic evolution of user relationships during propagation. Finally, the model's parameters depend on fitting specific datasets; enhancing its cross-platform generalization ability remains a key objective. In the future, we will integrate existing models and these improvements to more robustly predict the scale of malicious information propagation, thereby better managing online public opinion.

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